differentiated into foot, seta and capsule (Figure 2.22). The foot is small, conical and is embedded in the gametophyte. The seta

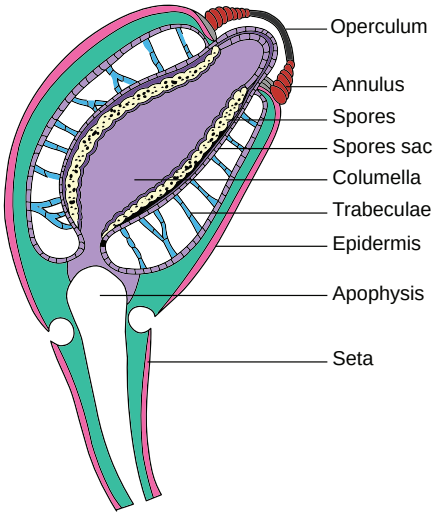


Figure 2.22: L.S. of capsule

is long, slender and conducts water and nutrients to the capsule . The capsule is differentiated into apophysis, theca proper and operculum. The cells constituting the wall of the capsule contain chloroplasts in them. The apophysis is the lowermost sterile region and connects the capsule with seta. The epidermis contains stomata which help in exchange of gases. The cells of the apophysis are photosynthetic, hence the sporophyte of *Funaria* partially depends on the gametophyte. The theca proper is the middle part and is fertile region of the capsule. It consists of a central columella surrounded by spore sac. The spore sac is surrounded by a single cylindrical air sac traversed by delicate filaments made up of parenchyma cells called **trabeculae**. The trabeculae extend from the outer wall of the spore sac to the innermost layer of the capsule wall. The spore sac contains spore mother cells which undergo meiotic division to produce haploid spores. The apical region consists of the operculum and peristome. The operculum is the lid of the capsule and comes out as a circular cup shaped lid after the dehiscence of the capsule. The peristome has one or two rows of thickened, tooth like projections found on the top of the capsule. They are hygroscopic and help in the dispersal of the spores. During favourble conditions the spores germinate to produce thread like green branched structure called protonema. It produces rhizoids and number of lateral buds which develop into new plants. In the life cycle of *Funaria* the haploid gametophytic phase (n)alternates with diploid sporophytic phase (2n) and shows alternation of generation (Figure 2.23).

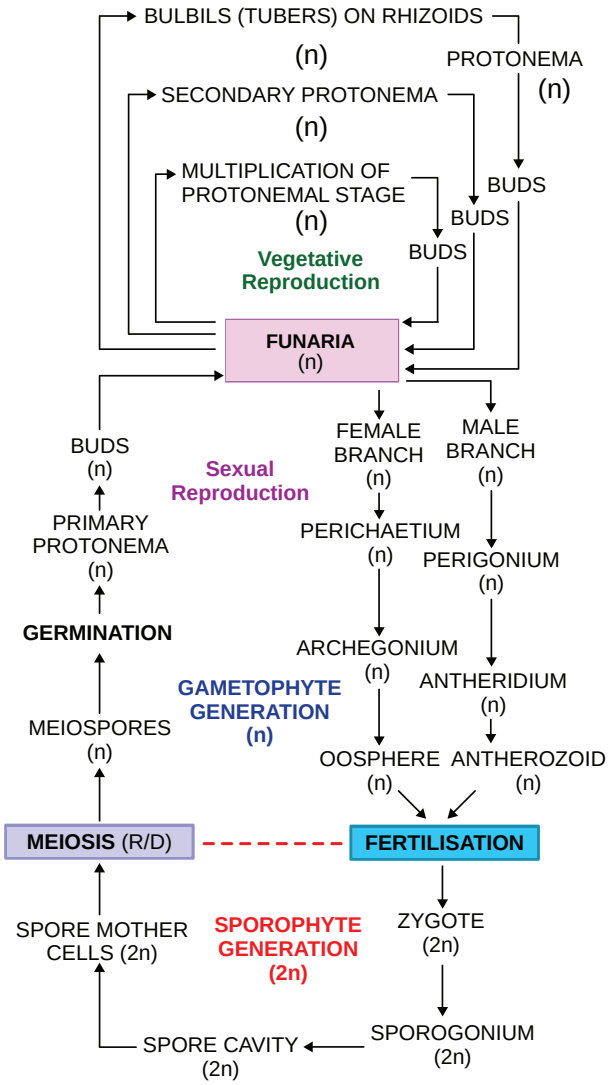


Figure 2.23: Life cycle of *Funaria*